tf.linspace Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/linspace

> *Generates evenly-spaced values in an interval along a given axis.*

Function Signatures

```
tf.linspace( start, stop, num, name=None, axis=0 )
```

Code Examples

```
tf.linspace(
start, stop, num, name=None, axis=0
)
```


```
tf.linspace(
start, stop, num, name=None, axis=0
)
```


```
tf.linspace(10.0, 12.0, 3, name="linspace") => [ 10.0  11.0  12.0]
```


```
tf.linspace(10.0, 12.0, 3, name="linspace") => [ 10.0  11.0  12.0]
```

```
tf.linspace([0., 5.], [10., 40.], 5, axis=0)
```

```
tf.linspace([0., 5.], [10., 40.], 5, axis=0)
```

```
array([[ 0.   ,  5.   ],
```

Documentation

Generates evenly-spaced values in an interval along a given axis.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.lin_space, tf.compat.v1.linspace

Used in the notebooks

- Multilayer perceptrons for digit recognition with Core APIs

- Advanced automatic differentiation
- Introduction to gradients and automatic differentiation
- Basic training loops
- TensorFlow basics
- Integrated gradients
- Learned data compression
- Basic regression: Predict fuel efficiency
- Scalable model compression
- TFP Release Notes notebook (0.12.1)

A sequence of num evenly-spaced values are generated beginning at start along a given axis.
If num > 1, the values in the sequence increase by
(stop - start) / (num - 1), so that the last one is exactly stop.
If num <= 0, ValueError is raised.

Matches
np.linspace's
behaviour
except when num == 0.

For example:

Start and stop can be tensors of arbitrary size:

Axis is where the values will be generated (the dimension in the returned tensor which corresponds to the axis will be equal to num)

Returns